

Global Supply Chain Performance Intelligence

1. Executive Summary

In the current analysis period, the supply chain network handled around **10,000 shipments** and generated total revenue of **\$1.40 billion**. The overall On-Time Delivery (OTIF) performance is **45.17%**, which is below the typical industry **benchmark of around 95%**, indicating that delivery reliability needs improvement.

The **average transit time is 6.02 days**, showing that shipments are generally moving within a reasonable time frame. **The average transportation cost is \$5.5 per KM**, which is currently under control but should be monitored as shipment volume grows.

From a financial point of view, the business is performing strongly with an overall **profit margin of 70%** and an **average revenue per shipment of about \$139.5K**. Supplier performance shows an **average lead time of 11.22 days**, which is acceptable but can be improved for better planning.

Focus Areas

- Improve on time delivery performance
- Reduce shipment delays in risky transport modes
- Improve supplier lead time reliability
- Monitor logistics cost efficiency
- Maintain strong shipment profitability

2. Business Problem Statement

The company operates a global supply chain network with multiple suppliers, warehouses, transport modes, and customer segments. Recently, management has faced challenges related to low on time delivery performance, rising transportation costs, and inconsistent supplier lead times. There is also a need for better visibility into warehouse activity and shipment profitability.

Currently, performance monitoring is limited and mostly reactive, making it difficult to quickly identify delays, high-cost routes, and operational risks. Therefore, the business requires an integrated analytics solution that can provide clear visibility into delivery performance, cost efficiency, supplier reliability, and overall supply chain health to support better data-driven decision-making.

3. Project Objective

The objective of this project is to build an end to end supply chain analytics solution that gives clear visibility into important operational and financial metrics. The solution helps business users monitor delivery performance, track logistics cost efficiency, evaluate supplier reliability, check warehouse and inventory health, and understand shipment profitability.

The project uses Python for data cleaning and feature creation, MySQL for storing and analyzing data, and Power BI for building interactive dashboards. The main goal is to support better and faster data-driven decision-making in the supply chain.

4. Methodology / Data Pipeline

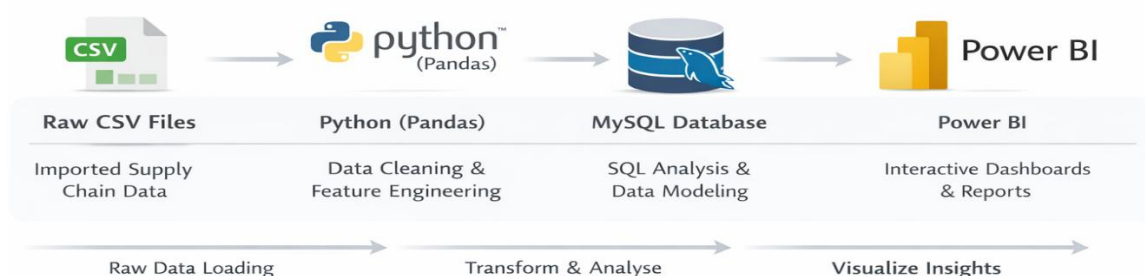
This project follows a clear end to end data pipeline to convert raw supply chain data into useful business insights.

First, raw CSV files were loaded into Python using Pandas. Basic data cleaning was performed, such as handling missing values, fixing data types, and standardizing columns. New features like Transit Time, Delay Flag, Fulfillment Speed, Cost per KM, and Profit Margin were created to support deeper analysis.

Next, the cleaned data was loaded into a MySQL database. Proper table relationships were created so that the data could be analyzed efficiently. SQL queries were then used to calculate important business metrics such as OTIF performance, delay risk by transport mode, customer revenue contribution, warehouse load ranking, and supplier lead-time analysis.

Finally, Power BI was connected to the MySQL database to build two interactive dashboards. The first dashboard focuses on logistics operations performance, and the second dashboard focuses on financial and supply chain health. These dashboards help business users monitor KPIs and make faster data-driven decisions.

Data Pipeline: From Raw Data to Insights



5. Key Insights

1. Overall Delivery Performance (OTIF)

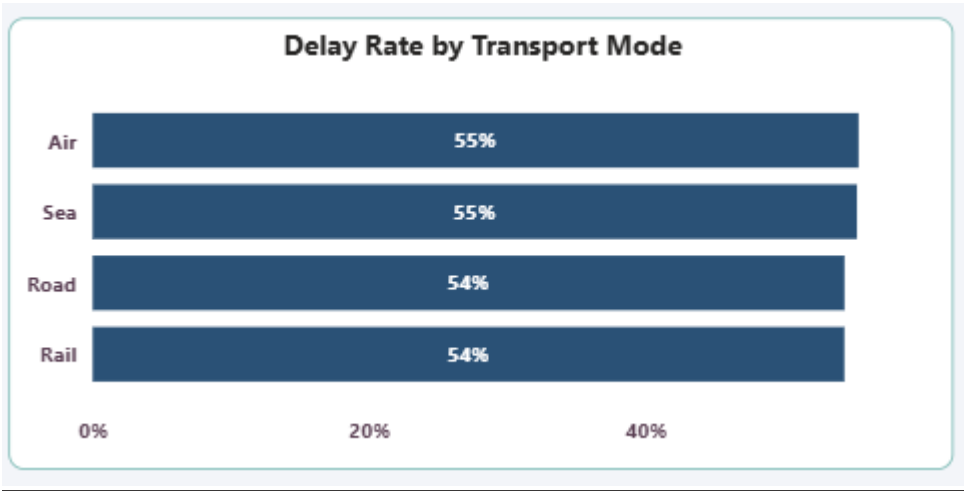
OTIF (On-Time In-Full) shows the percentage of shipments delivered on time. In this analysis, the OTIF is 45.17%, which is much lower than the industry benchmark of around 95%.

This means many shipments are getting delayed, which can affect customer satisfaction. The business should focus on reducing delays and improving delivery reliability.

2. Delay Risk by Transport Mode

The analysis compares shipment delay percentage across different transport modes. The delay rate is relatively similar across Air (55%), Sea (55%), Road (54%), and Rail (54%).

This indicates that delivery delays are not limited to a single transport mode but are occurring across the network. Therefore, the business should focus on overall logistics process improvement rather than targeting only one transport mode.



3. Slowest Logistics Routes

This analysis highlights the routes with the highest average transit time across regions. The data shows that shipments to North America generally take slightly longer compared to APAC and Europe routes.

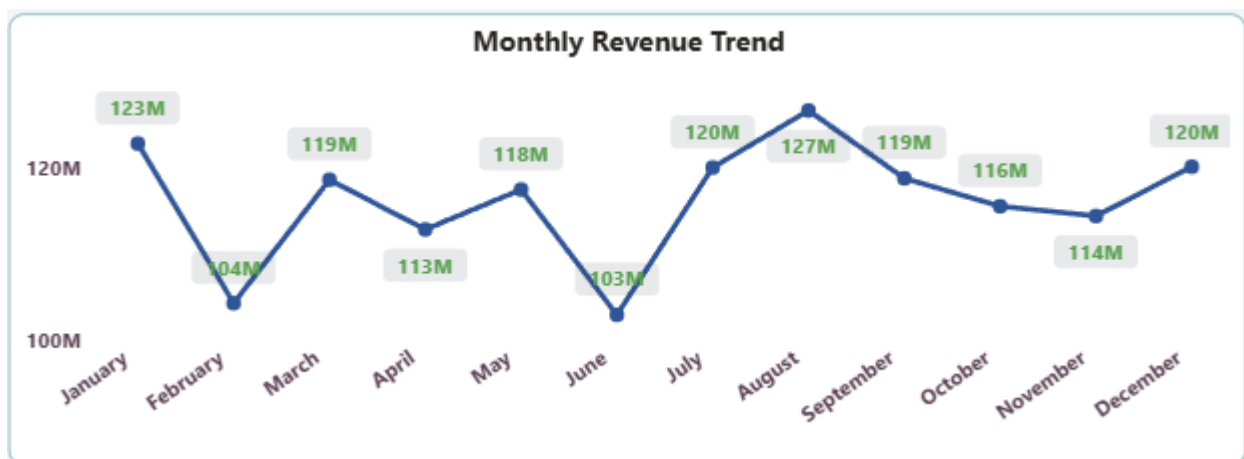
Although the differences are not very large, these slower lanes can still increase the risk of late deliveries if shipment volume grows. The business should continue monitoring these routes and explore opportunities such as route optimization or carrier improvement to reduce transit time.

Slowest Logistics Routes			
Country	APAC	Europe	North America
Germany	5.94	5.91	5.99
India	6.04	5.95	6.06
Singapore	5.98	6.03	6.07
Uae	5.96	6.23	6.03
Usa	5.87	6.16	6.21

4. Monthly Revenue Trend

The monthly revenue trend shows that revenue remains generally stable throughout the year, with normal ups and downs. Revenue drops slightly in February (~104M) and June (~103M), but quickly recovers in the following months.

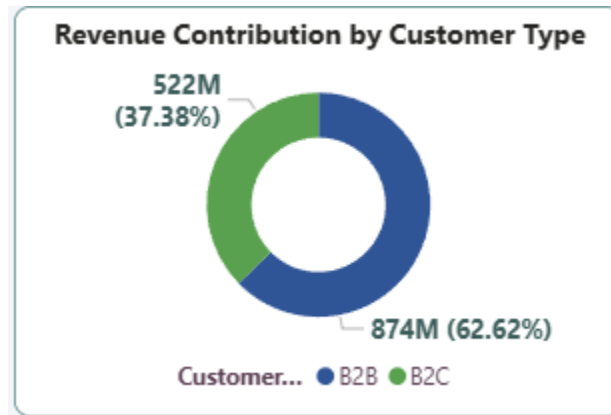
The highest revenue is observed around August (~127M), indicating strong demand during that period. Overall, the business is maintaining steady revenue performance, but it should continue monitoring low-performing months to avoid future decline.



5. Revenue Contribution by Customer Type

The analysis shows that **B2B customers generate the majority of revenue (about 62.6%)**, while **B2C contributes around 37.4%**.

This indicates that the business is currently more dependent on B2B customers for revenue. While this is positive for bulk sales stability, the company can also explore opportunities to grow the B2C segment to achieve a more balanced customer mix.



6. Supplier Lead Time Performance

This analysis shows the average lead time taken by suppliers to deliver purchase orders. The average supplier lead time is around **11.22 days**, which indicates moderate procurement efficiency.

However, some suppliers show slightly higher lead times, which may affect replenishment planning and inventory availability. The business should closely monitor supplier performance and work with slower vendors to improve delivery consistency

7. Shipment Profitability Health

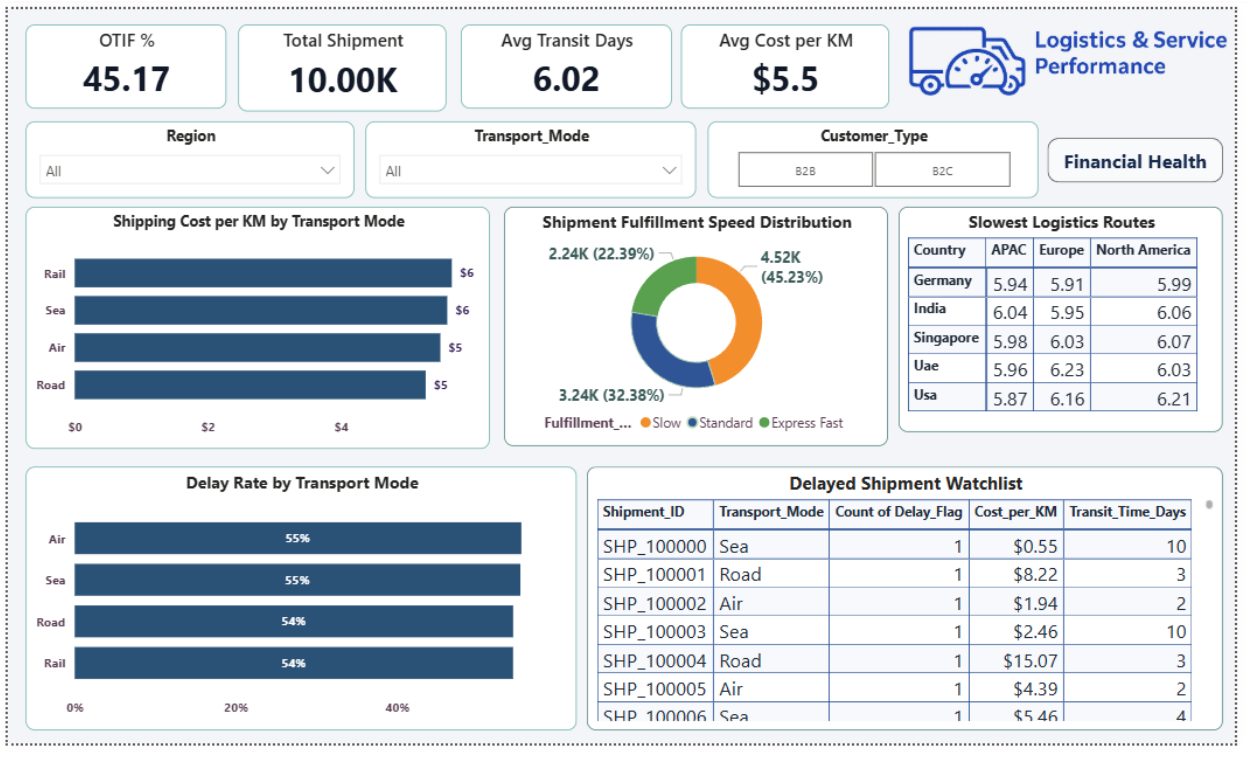
The analysis shows that the overall profit margin is around **70%**, which indicates strong shipment-level profitability. This means the business is currently managing its revenue and logistics costs effectively.

However, it is important to continuously monitor cost per KM and transportation efficiency to ensure that profit margins remain stable as shipment volume increases

6. Dashboard Overview

To provide clear and actionable visibility into supply chain performance, two interactive dashboards were developed in Power BI. These dashboards allow business users to monitor key metrics, identify risks, and support data-driven decision-making.

Logistics & Service Performance focuses on operational efficiency and delivery reliability. It tracks key metrics such as OTIF performance, average transit time, cost per KM, delay rate by transport mode, and fulfillment speed distribution. This dashboard helps operations teams quickly identify delay risks, slow logistics routes, and transportation performance issues.



Financial & Supply Chain Health focuses on financial performance and supply stability. It provides visibility into total revenue, overall profit margin, supplier lead time performance, revenue contribution by customer type, warehouse load utilization, and monthly revenue trends. This dashboard supports finance, procurement, and planning teams in monitoring profitability and supplier efficiency.



Business Recommendations

1. Improve OTIF Performance

Current OTIF is below the desired benchmark. Implement weekly tracking, investigate delay root causes (transport inefficiencies, warehouse dispatch), and target 90%+ OTIF to improve service reliability.

2. Optimize Transport Mode Usage

Since delay risk is similar across modes, focus on route-level optimization and carrier performance review. Use air/road selectively for urgent shipments and prioritize sea/rail for cost efficiency.

3. Address Slow Logistics Lanes

High transit times in specific origin destination lanes require network redesign, alternate routing, or carrier renegotiation. Continuously monitor lane KPIs to reduce variability.

4. Strengthen Supplier Lead-Time Control

Lead-time variability impacts planning. Develop a supplier scorecard, review underperforming vendors, and consider contract renegotiation or secondary sourcing for risk mitigation.

5. Diversify Revenue Mix

B2B drives the majority of revenue. Continue strengthening B2B while expanding B2C opportunities to reduce dependency risk and improve revenue quality.

6. Balance Warehouse Utilization

Uneven warehouse loads indicate capacity risk. Optimize inventory allocation and capacity planning, and closely monitor high-load facilities during peak periods.

8. Project Conclusion

This project successfully developed an end-to-end supply chain analytics solution using Python, MySQL, and Power BI. The analysis provided clear visibility into delivery performance, logistics cost efficiency, supplier reliability, and overall financial health of the supply chain.

The dashboards highlight key improvement areas, especially in on-time delivery performance and supplier consistency, while also confirming strong shipment profitability. By implementing the recommended actions and continuously monitoring KPIs through automated dashboards, the organization can improve operational efficiency and support better data-driven decision-making.